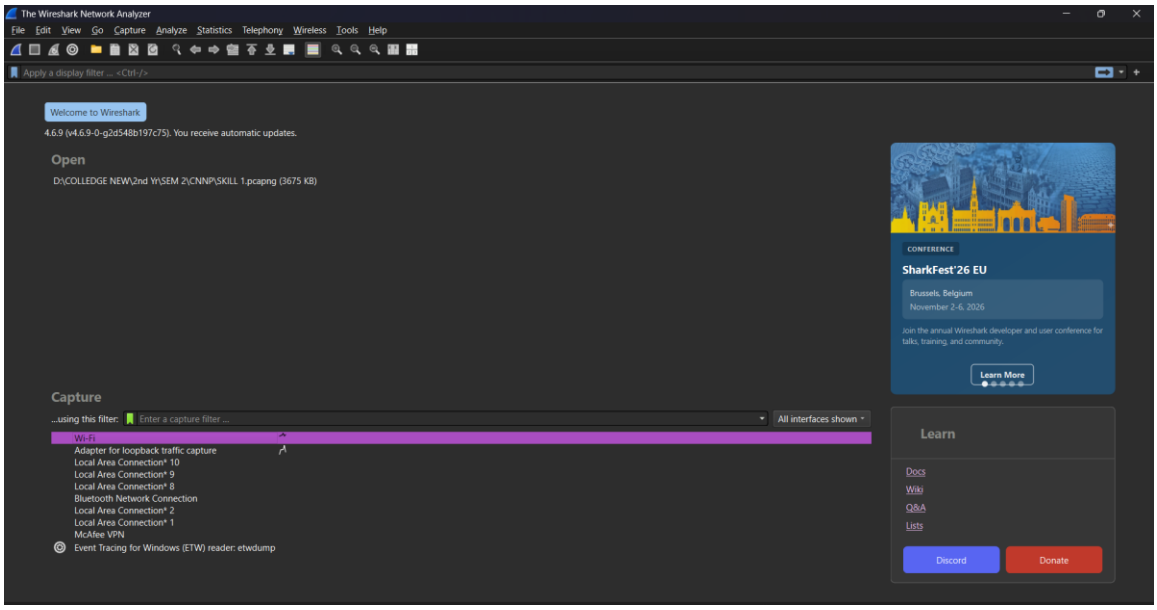
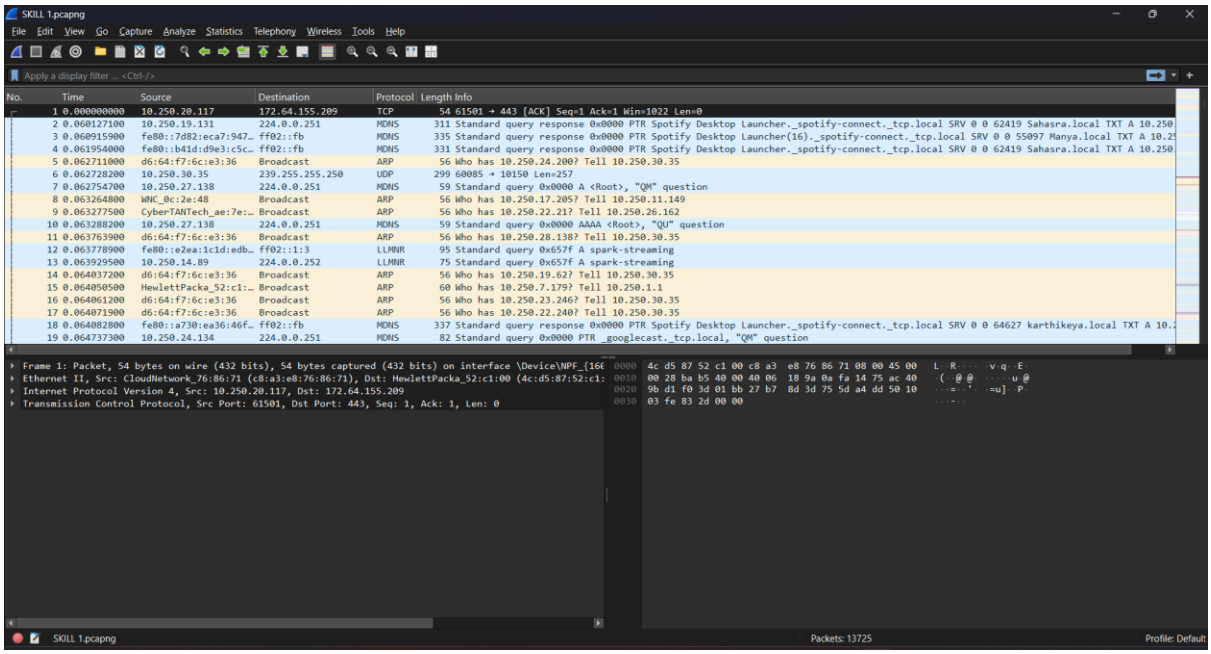
SKILL-1

NAME: Salihundam Svara Haasini

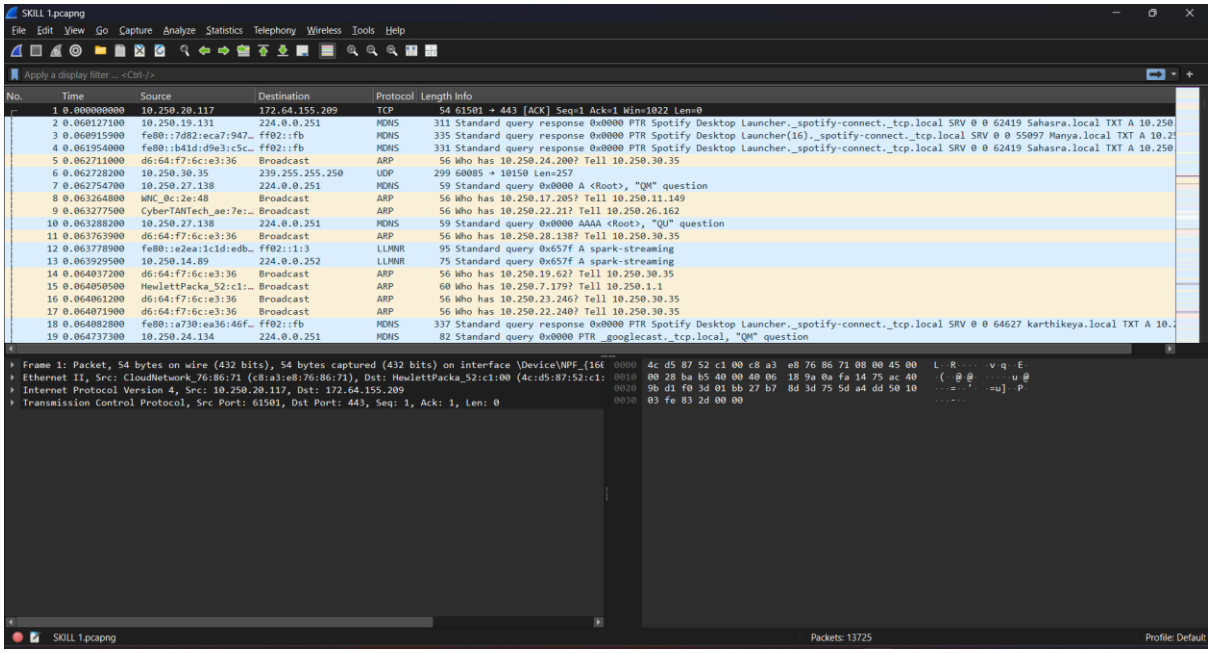
Step 1:



STEP 2

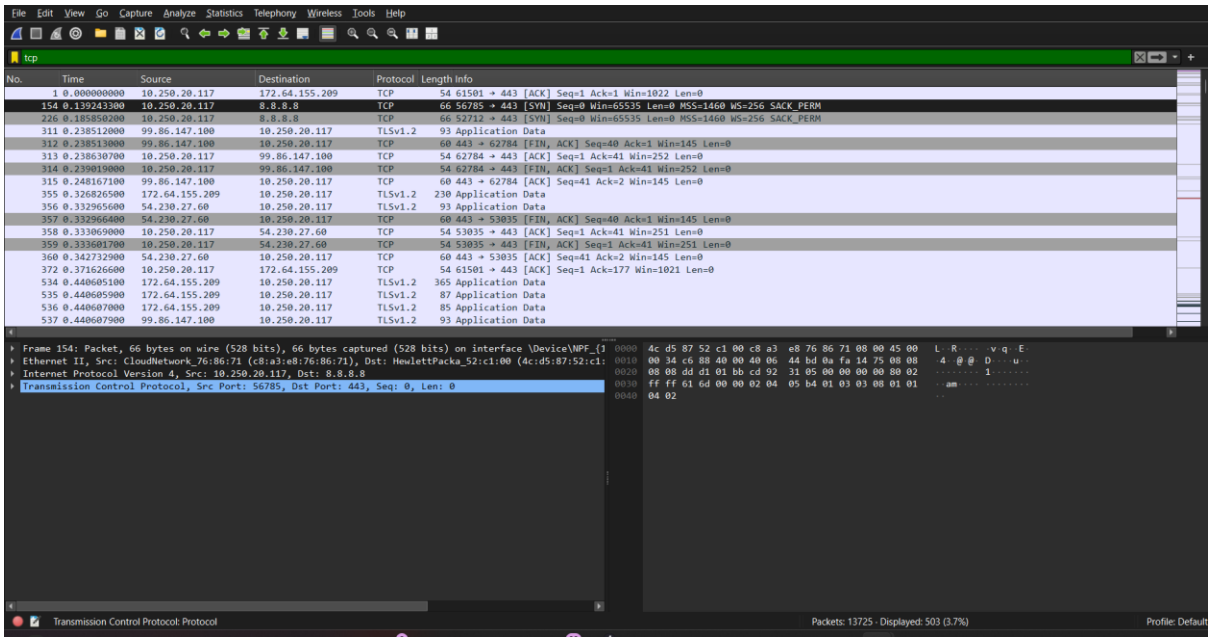


STEP 3 :

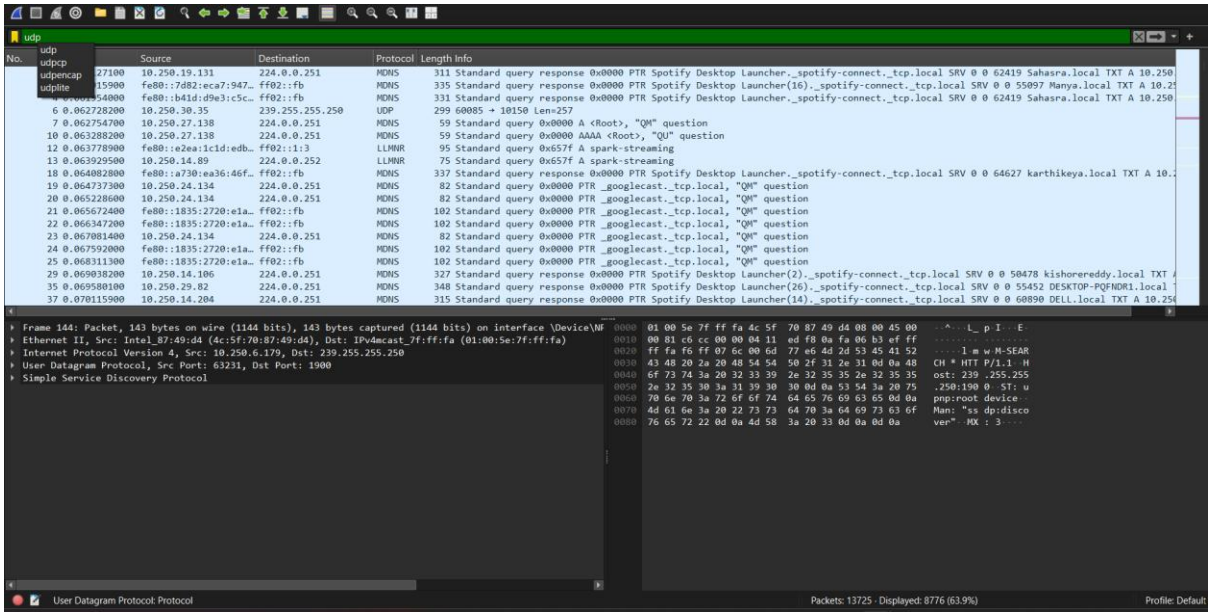


STEP 4: Apply display filters

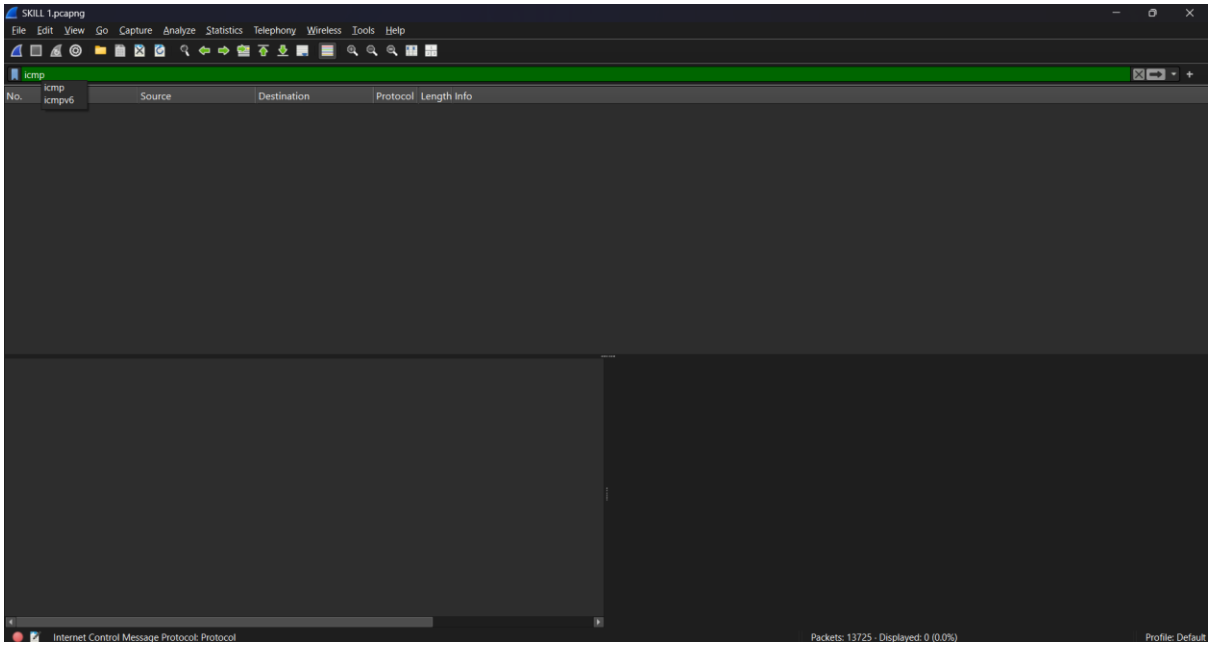
TCP:



UDP:

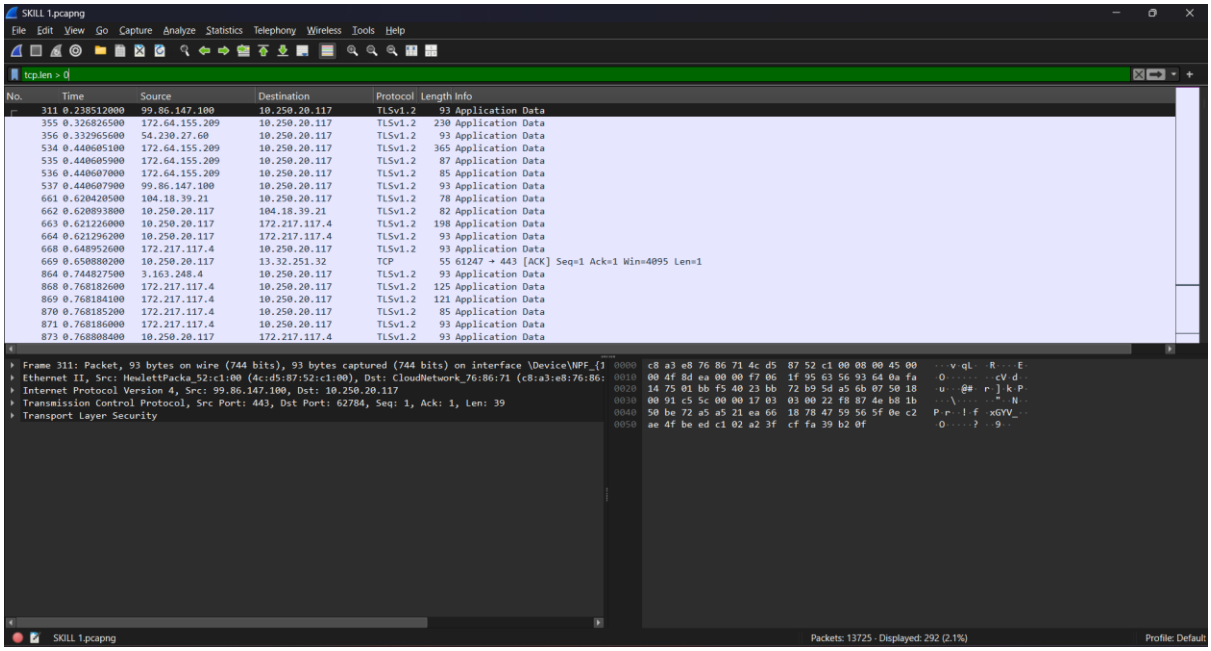


ICMP

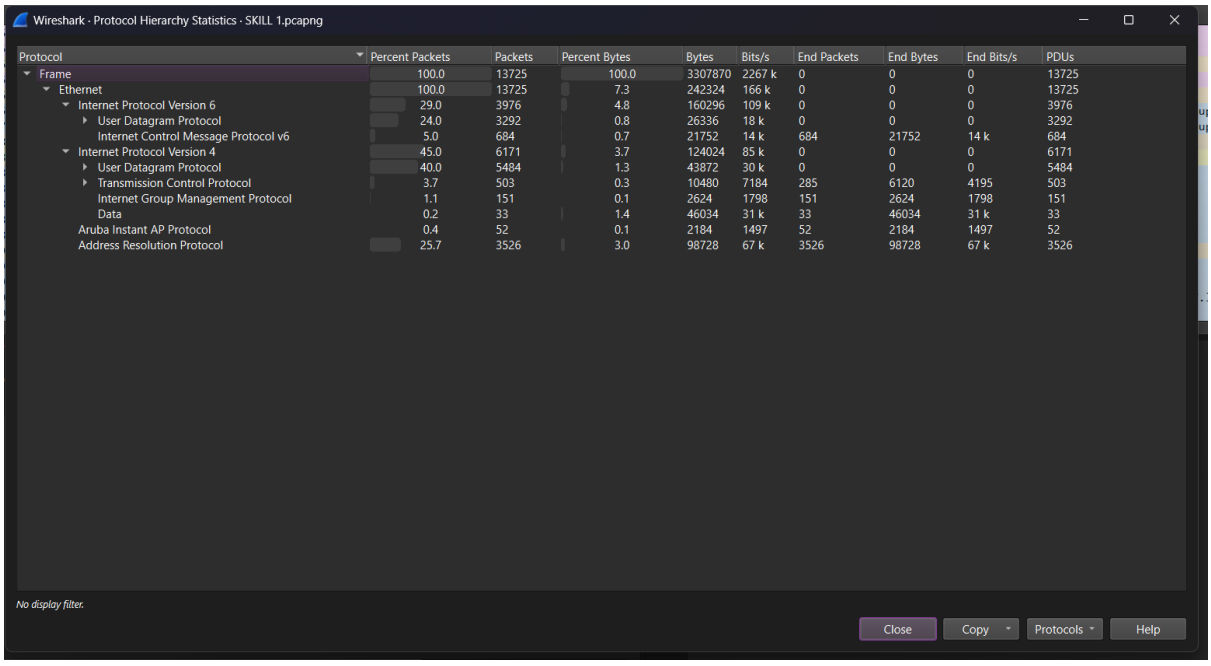


Observation: No ICMP packets were found in the captured traffic.

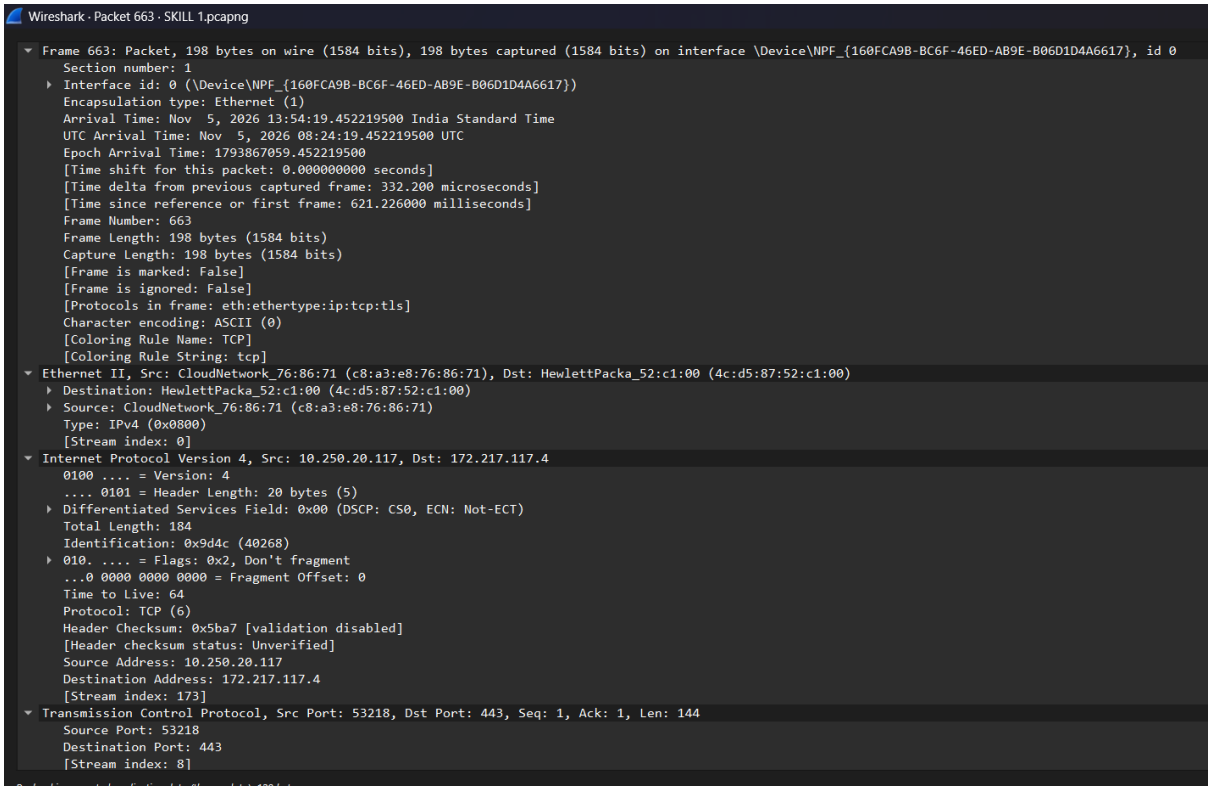
Tcp.len >0

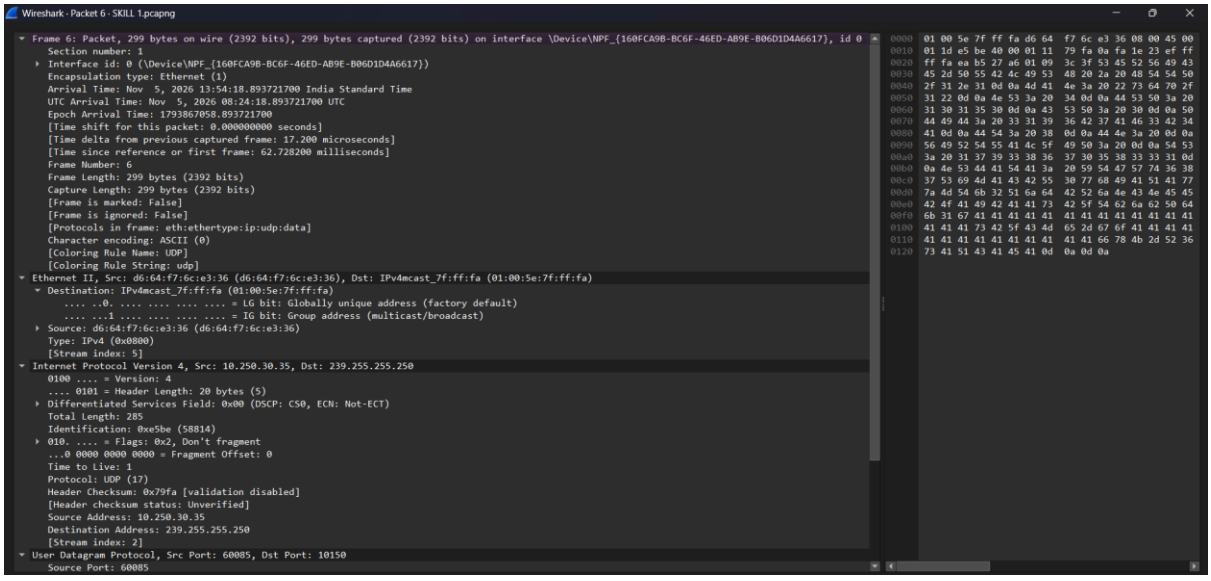


Protocol hierarchy



STEP 5 : TCP Header Inspection and UDP Header Inspection.





STEP 6 :

TCP packet — Frame 663

Given:

- Ethernet Header = 14 bytes
- IPv4 Header = 20 bytes
- TCP Header = 20 bytes
- TCP Payload = 144 bytes

Total Overhead:

$$14 + 20 + 20 = \boxed{54 \text{ bytes}}$$

Payload Efficiency:

$$\frac{144}{198} \times 100 = \boxed{72.73\%}$$

UDP packet — Frame 6

Given:

- Ethernet Header = 14 bytes
- IPv4 Header = 20 bytes
- UDP Header = 8 bytes
- UDP Payload = 257 bytes

Total Overhead:

$$14 + 20 + 8 = \boxed{42 \text{ bytes}}$$

Payload Efficiency:

$$\frac{257}{299} \times 100 = \boxed{85.95\%}$$

STEP 7: OBSERVATION TABLE

Parameter	TCP Packet (Frame 663)	UDP Packet (Frame 6)

Frame Length	198 bytes	299 bytes
Ethernet Header Length	14 bytes	14 bytes
IP Header Length	20 bytes	20 bytes
Transport Header Length	20 bytes	8 bytes
IP Total Length	184 bytes	285 bytes
Transport Header Length	20 bytes	8 bytes
Payload Length	144 bytes	257 bytes

HEADER OVERHEAD AND EFFICIENCY

Parameter	**TCP Packet**	**UDP Packet**
Total Header Overhead	54 bytes	42 bytes
Payload Length	144 bytes	257 bytes
Total Frame Length	198 bytes	299 bytes
Payload Efficiency	72.73%	85.95%

QUESTIONS AND ANSWERS

1. What is encapsulation?

Encapsulation is the process of adding protocol headers to data as it passes through different network layers before transmission.

2. What is the purpose of the Ethernet header?

The Ethernet header contains the source and destination MAC addresses and identifies the type of data being carried. It helps deliver frames across a local network.

3. What information is stored in the IP header?

The IP header contains the source and destination IP addresses, header length, total length, TTL (Time to Live), protocol information, and other fields required for packet delivery.

4. What is the difference between TCP and UDP headers?

A TCP header is at least 20 bytes and contains sequence numbers, acknowledgement numbers, and control flags for reliable communication. A UDP header is only 8 bytes and supports simpler communication without guaranteed delivery.

5. Why does each protocol add a header?

Each protocol adds a header to provide the control information needed for its specific function, such as addressing, routing, error detection, and reliable data delivery.

6. What is packet overhead?

Packet overhead refers to the extra bytes added by protocol headers and other control information during transmission, apart from the actual application data.

7. How does header overhead affect network efficiency?

Header overhead reduces the proportion of transmitted data available for actual application information. Higher overhead relative to payload results in lower network efficiency, especially when transmitting smaller packets.